AcademicNotesAI Fused Dataset – EDA & Dashboard Report

Data Visualization assignment · Data Science & Analysis · Tools: Python, Pandas, Plotly, Dash (VS Code)

1. Objective

This report explains the exploratory data analysis (EDA) and the interactive dashboard built for the **AcademicNotesAI fused dataset**. AcademicNotesAI is an LLM-powered study assistant where students chat with an AI, search YouTube tutorials, upload diagrams for explanation, take quizzes and save notes. The goal is to understand *who uses the app, when, for which topics, and how well they perform*, and to present those answers through 12 different chart types with filters.

2. Dataset Description

The workbook has **75 rows × 40 columns** on a single sheet ("Fused Dataset"). It was created by fusing seven JSON exports into one table; the column `record_type` tells which source a row came from. Because every source has different fields, most columns are filled only for some record types.

Record type	Rows	What it contains	Fields that are filled
chat_history	27	Questions asked to the AI chat	chat_title, message_count
youtube_history	12	YouTube tutorial searches	topic, results_count (always 6)
image_study_history	9	Diagrams / charts uploaded for explanation	image_name, instruction
users	8	Account sign-ups	user_name, email, created_at
favourites	8	Saved notes	question, answer, sources, lengths
activity_history	6	Chat activity log	description, topic

Record type	Rows	What it contains	Fields that are filled
quiz_history	5	Quiz attempts	topic, difficulty, score, total_questions, percentage

The period covered is **1 Aug 2026 – 14 Oct 2026**, with 9 user accounts that belong to 4 named people.

3. Data Cleaning & Feature Engineering

The dataset is small and has no duplicate rows or duplicate `record_id` values, but the fusion created several issues that had to be handled before any chart was drawn.

- **Placeholder text.** Missing values are stored as the string "Not Applicable" (or 0 for numbers) rather than as real blanks. Eighteen columns are more than 85 % empty, which is a structural effect of fusing different sources, not a data-quality error.
- **Constant columns.** `email_verified` and `disabled` are 0 in all 75 rows, so they carry no information and were left out of every chart.
- **Three date formats.** Dates appear as `YYYY-MM-DD`, `YYYY-MM-DD HH:MM` and `YYYY-MM-DD HH:MM:SS UTC`. They were parsed with `format="mixed"` into one datetime column.
- **Unreliable hour.** `activity_hour` is 0 for all 27 chat_history rows because those rows store only a date. Hour-of-day charts therefore use only the 48 rows that carry a real clock time.
- **User identity.** One person can own several UIDs (Prajval has 3, Bhakti 2, Pranjal 2). A `user` column was built by mapping each UID to the name found in the users records; the 35 rows (27 chats and 8 saved notes) whose UID has no sign-up record are labelled "Unlinked account".
- **Engineered features.** `topic_group` (a keyword classifier over all text fields that turns messy titles into 9 subjects plus "General"), `prompt_text`, `prompt_words`, `prompt_chars`, `weekday`, `week`, `month`, `day_part`, and a readable `activity` label.

4. Key EDA Findings

Activity mix. AI chat is the dominant feature: 33 of 75 records (44 %) when chat_history and activity_history are combined, followed by YouTube searches (12), image study (9), saved notes (8), sign-ups (8) and quizzes (5).

Topics. Machine Learning and Cloud Computing are tied at 12 records each, then AI & LLMs (9), Python (6), Data Science (6), DBMS (5), Operating Systems (5), Data Structures (3) and Cyber Security (2). Fifteen records (sign-ups and unnamed image uploads) have no topic.

Time. 70 of 75 records fall in August and 5 in October; there are no records in September. Saturday is the busiest weekday (28 records), then Tuesday (18); Friday has only 1. Among rows with a clock time, 19 fall in the morning (06–11 h), 16 at night (00–05 h), 10 in the afternoon and only 3 in the evening, so the typical session is an early-morning study session.

Users. Payal Pramod Pawar is the heaviest named user with 26 records (10 YouTube searches, 7 image uploads, 5 chats, 3 quizzes, 1 sign-up). Bhakti has 9 records. Prajval and Pranjal only appear as sign-ups.

Chat behaviour. 25 of 27 chats contain exactly 2 messages (one question and one answer); the maximum is 8 messages for "Explain Cloud Security". Several prompts are repeated: "what is AI", "what is machine learning" and "explain cloud security" each appear 4 times.

Quizzes. The average score is **72 %** and 3 of 5 quizzes passed the 60 % mark. Both Easy quizzes scored 100 %, while Medium quizzes averaged only 53 % (40 %, 80 % and 40 %). The weakest result was the 10-question Python quiz at 40 %.

Saved notes. Answers are long: mean 1,938 characters (range 1,139–2,798) for questions averaging 27 characters. Inside the 8 notes the question-answer length correlation is weak (r = 0.24).

Prompt length. Chat prompts average 4.3 words, saved-note questions 4.1, image instructions 3.4, YouTube searches 2.8 and quiz topics 1.4 words. Users write very short prompts everywhere.

5. Dashboard Design

Layout. A header, a sticky filter bar, six KPI cards (total records, user accounts, AI chats, average quiz score, average note length, top topic) and four sections: Overview, Content & Topics, Distributions & Behaviour, and Relationships & Performance. Each chart has a one-line interpretation note under it.

Filters. Date range, User, Activity type and Topic. All 12 charts and all KPI cards update together, and a Reset button restores the full dataset. Empty selections show a friendly "No data for the selected filters" message instead of an error.

Colour system. Each activity keeps the same colour in every chart so the reader can follow it across the page: AI Chat indigo `#4F46E5`, YouTube red `#EF4444`, Image Study amber `#F59E0B`, Quiz green `#10B981`, Saved Note cyan `#06B6D4`, Sign-up grey `#94A3B8`. Quiz difficulty uses a traffic-light scheme (green easy, amber medium, red hard). Magnitude charts (heatmap, polar bars) use a single-hue indigo scale, and the correlation matrix uses a diverging red–blue scale centred on zero. The page uses a light background with white cards so colour is reserved for data.

6. The 12 Visualizations

#	Chart	Type	Why this type	What it shows
1	Activity Mix	Donut	Part-to-whole of a few categories	AI chat is the largest share; quizzes the smallest
2	Weekly Activity Trend	Stacked area	Time trend plus composition	Usage comes in bursts, with a gap in September
3	Weekday × Activity	Heatmap	Two categorical dimensions	Saturday and Tuesday dominate; Friday is nearly empty
4	Records per User	Stacked bar	Compare totals and mix	Payal is the power user; others mostly only signed up
5	Most Studied Topics	Horizontal bar	Ranking with long labels	Machine Learning and Cloud Computing lead
6	Activity → Topic	Sunburst	Hierarchy (feature, then topic)	Chat covers most topics; quizzes spread thinly
7	Prompt Length by Activity	Box plot with points	Distribution, median and outliers	Short prompts everywhere; chat has the widest spread
8	Time-of-Day Usage	Polar bar (clock)	Cyclic variable (24 h)	Peaks around 04–05 h and 09–12 h
9	Question vs Answer Length	Scatter with trend line	Relationship of two numbers	Weak positive link; answers are always long

#	Chart	Type	Why this type	What it shows
10	Quiz Scores	Bar with pass line	Compare against a threshold	Easy 100 %, Medium 53 % on average
11	Event Timeline	Strip / scatter over time	Individual events per user	Shows when each person was active
12	Correlation Matrix	Heatmap (diverging)	Many pairwise relationships	Score, total questions and percentage move together

7. Main Insights

- Students use the app mainly as a **question-answering tutor**: chat is the top feature, and most chats are a single question and answer.
- **Machine Learning, Cloud Computing and AI** are the most requested subjects, which matches a data-science curriculum.
- Engagement is **bursty and weekend or early-morning heavy**, with a complete gap in September.
- **Harder quizzes expose weaknesses.** Medium quizzes score about 47 points lower than Easy ones, so adaptive practice or revision prompts would help.
- Prompts average only 3–4 words, so the LLM often works from very little context; suggested follow-up questions could improve answers.
- Repeated identical questions suggest that **caching or a FAQ answer** for common topics would save cost.

8. Limitations

The dataset has only 75 rows, and quiz results come from just 5 attempts and saved-note statistics from 8 notes, so every pattern here is **indicative, not statistically conclusive**. Correlations computed over the whole fused table are inflated by the structural zeros: for example, question and answer length correlate at 0.89 across all rows but only 0.24 inside the saved notes. The chat records contain no time of day, no answer text and no timing, which limits behavioural analysis. Topic groups come from a keyword classifier and may misclassify unusual titles.

9. Conclusion

The fused dataset was cleaned, enriched with derived features and explored through descriptive statistics, distributions, relationships and time analysis. The resulting Dash dashboard presents 12 different chart types with four linked filters, a consistent colour system and short interpretations, so a reader can explore usage patterns of the AcademicNotesAI platform without writing code.

10. How to Run the Project

1. Unzip the project folder and open it in VS Code.
2. Create a virtual environment and run `pip install -r requirements.txt`.
3. Run `python eda.py` to print the full EDA in the terminal.
4. Run `python dashboard.py` and open `http://127.0.0.1:8050` in a browser.
5. Optionally run `python export_figures.py` to save every chart as a PNG for slides or screenshots.